

- 3 A program stores integers in ascending order in an ordered binary tree. The tree is implemented as a 2D array.

Each node is stored with three values:

- a pointer to the left node
- the data
- a pointer to the right node.

All null values are stored as `-1`

The binary tree can store up to 50 nodes.

Nodes cannot be deleted from the binary tree.

- (a) The binary tree is stored as a global array with the identifier `TreeArray`. The left pointer, the data and the right pointer of each node are initialised to `-1`

The global variable `RootPointer` stores the index of the root node in the tree, initialised to `-1`

The global variable `FreeNode` stores the index of the next free node in the array, initialised to `0`

Write program code to declare and initialise `TreeArray`, `RootPointer` and `FreeNode`

Save your program as **Question3_N25**.

Copy and paste the program code into part **3(a)** in the evidence document.

[3]

- (b) The procedure `AddNode()`:

- takes an integer to store in the binary tree as a parameter
- stores the parameter in the next free node in the array
- finds the position to store the data in the tree by following the appropriate pointers
- updates the pointer of the new node's parent node.

The procedure outputs "The tree is full" if the parameter cannot be stored because the tree is full.

Write program code for `AddNode()`

Save your program.

Copy and paste the program code into part **3(b)** in the evidence document.

[7]